

Happy Yadav

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About Me

I am an Applied AI Engineer with hands-on production experience in building real-time VoiceAI systems, multi-channel conversational agents, and LLM-powered backend pipelines. My core strength lies in architecting end-to-end AI systems from low-latency V2V (voice-to-voice) streaming pipelines using LiveKit/pipecat and WebRTC, to LangChain-orchestrated chatbot services with MCP-based enterprise tool integrations. On the full-stack side, I build scalable backend services using Python (FastAPI) and frontend interfaces using React. I specialize in agentic AI systems, RAG pipelines, multi-provider LLM routing, and real-time communication stacks, and have delivered these in production across web, WhatsApp, and telephony channels.

Skills

Languages: Python, JavaScript, TypeScript, C++, SQL

Frameworks & Libraries: FastAPI, React, Node.js, Express.js, Redux, Streamlit, Selenium, BeautifulSoup

Tools: Git, Docker, Google Cloud, Firebase, LiteLLM, UV, Husky, Ngrok, Cloudflare

Databases: PostgreSQL, MongoDB, Redis, SQLite, Firestore

AI & ML: Open Fine-Grained Authorization (OFGA), LangChain, LangGraph, LangSmith, Retrieval Augmented Generation (RAG), LLM Embeddings, Prompt Engineering, MCP (Model Context Protocol), UCP (Universal Control Protocol), Pinecone, FAISS, litellm

Voice & Real-Time: LiveKit (WebRTC), Automatic Speech Recognition (ASR), WebSocket Audio Streaming, Voice Activity Detection (VAD), Speaker Diarization, STT/TTS Voice pipeline, DeepFilterNet, Twilio, Exotel, μ -law, FFmpeg, pipecat

Others: REST API Integrations, Microservice and Monolithic Architecture, Async Pipelines, CI/CD

Professional Experience

Applied AI Engineer - EaseMyTrip.com

Gurugram, Haryana, India

Aug 2025 – Present

Key Projects:

Multipurpose Real-Time Voice AI Agent

Roles & Responsibilities:

- Built one channel-agnostic voice pipeline — STT → turn detection → LLM → TTS — that serves both PSTN calls and an in-browser voice widget, with custom wire-format serializers (μ -law/PCM 8 kHz) for Acefone plus a WebRTC path for the web. Integrated Deepgram Nova-3 STT, Groq LLM and pluggable TTS (Murf/Gemini/Sarvam/OpenAI) behind a single provider-factory abstraction, tuned to a sub-900 ms end-to-end turn budget with per-stage STT/LLM/TTS latency instrumentation.
- Engineered natural-conversation behaviour: sub-200 ms barge-in with a minimum-word threshold to reject background speech, contextual hold phrases when the LLM is slow, sequence watermarking so a stale reply is never spoken, and turn-fragment stitching so a request split across turns isn't re-asked. Added a guardrail processor chain (regex-based tool-call/PII/markup muting, duplicate-speech suppression, Hindi digit and Devanagari normalization) and procedurally generated call-center ambience, delivering a bilingual Hindi-English agent that reads as human on a live line.
- Designed a structured voice_action contract letting the LLM drive real telephony mid-call warm transfer to a human agent, language switch, hang-up gated on TTS playback completion so the action fires only after the bot finishes speaking. Built the supporting ops layer: WebRTC playground for auditioning voices/sample rates with live latency charts, per-turn call records and dispositions, and structured JSON call logging with PII redaction.

EMT Chatbot - Multi-Channel Conversational AI

Roles & Responsibilities:

- Built the core chatbot service powering both the EaseMyTrip website and WhatsApp bot with persistent multi-turn context via PostgreSQL-backed JSONB chat history and Redis session caching.
- Implemented an LLM Factory pattern supporting dynamic provider switching between Groq and OpenAI with

configurable temperature and model selection for cost-aware inference routing.

- Developed an optional-parameter extraction sub-chain that enriches tool calls with user-stated preferences without hallucinating unspecified values, significantly improving tool invocation accuracy. Also designed channel-aware system prompt management (Website vs. WhatsApp) with time-context injection to ensure temporally accurate, channel-optimized AI responses.

WhatsApp Travel Bot

Roles & Responsibilities:

- Engineered a production WhatsApp bot integrated with the EMT chatbot service via Meta Webhook APIs, handling realtime message routing, session management, and structured payload rendering for travel queries.
- Built Redis-backed session storage with configurable TTLs (session: 1 hr, pending data: 30 min) and a modular message-builder pipeline generating WhatsApp-compatible payloads (text, lists, interactive buttons). Also added the state change logic to switch between AI flow and Non AI flow based on user's query along with realtime customer agent assistance to user.
- Collaborated with Infra team to deploy via Docker with a containerized Redis service; integrated dev-tunnel support for local webhook testing against the Meta developer platform.

EMT Tools Ecosystem - MCP Tool-Calling Framework

Roles & Responsibilities:

- Contributed to the MCP-based tool ecosystem enabling dynamic enterprise API invocations: flight search, hotel lookup, train/bus booking, post and pre booking operations, and OTP-based login.
- Designed channel-aware tool filtering (WhatsApp-specific exclusions) and an intent-classification guard layer to handle unsupported query types with graceful main-menu fallback flows.
- Implemented a two-stage tool execution pipeline: primary LLM tool-call chain for parameter extraction, followed by an optional-params enrichment sub-chain for precision-enhanced API invocations.

Easymail AI Automation

Roles & Responsibilities:

- Built an agentic email-drafting engine that auto-generates grounded customer-support replies, replacing manual template hunting for 21 departments and 434 live templates. Architected a 10-stage LangGraph pipeline (intent detection → booking enrichment → template selection → draft → QA) with a confidence-gated retry loop a strict LLM scorer grades each draft 0–100 and regenerates with targeted feedback below an 85 threshold, hard-capped at 3 attempts for bounded latency and cost.
- Engineered privacy-safe, zero-leak LLM processing using Microsoft Presidio with a reversible masking layer names, phones, booking IDs, PNRs, cards and multi-currency amounts are swapped for named placeholders before *any* prompt, with a single per-request masker keeping the map consistent across query, OCR text and enriched API data, then restored post-scoring. Added hybrid attachment OCR (PyMuPDF text layer + RapidOCR ONNX fallback) so scanned tickets and screenshots become usable draft context, with per-file soft-failure and concurrent fetch.
- Designed a self-improving feedback loop and O(1)-cost template routing that scale with catalogue size. BM25 pre-ranking over an LLM-generated template manifest keeps prompt size flat even for the 130-template Care department; a Postgres metadata store caches every run so recreate/polish need one LLM call and zero re-fetches of booking/OCR/KB APIs. Agent feedback is injected immediately, then consolidated into a single per-department "learning" at batch threshold bounding token spend while compounding quality. Shipped fully containerised (Docker Compose) with graceful degradation at every dependency (mock LLM, regex PII, optional KB/persistence).

EaseMyTrip ChatGPT App

Roles & Responsibilities:

- Contributed to launching EaseMyTrip's MCP-powered ChatGPT App by exposing enterprise travel APIs as production-ready MCP tools for conversational AI interactions within ChatGPT.
- Worked on secure MCP tool integration, parameter extraction pipelines, and channel-aware conversational

workflows to support reliable enterprise-grade travel assistance like hotel, flight and train discovery inside ChatGPT.

Backend Developer Intern - Hunar.ai
Gurugram, India

Jun 2025 – Aug 2025

- Designed and optimized PostgreSQL query pipelines for data extraction workflows serving 20+ enterprise clients with 100,000+ records, improving reporting efficiency by 30%.
- Built RESTful APIs and contributed to schema optimization reducing query execution time across core reporting modules.
- Explored LLM integration patterns and contributed to backend modules supporting AI-assisted features within the platform.

Technical Lead - Coding Ninjas Club, Chitkara University
Punjab, India

2024 – 2025

- Led technical workshops and mentored 100+ club members on AI/ML, full-stack development, and competitive programming.
- Organized hackathons, coding sprints, and project showcases, fostering a culture of applied engineering and product thinking.
- Guided juniors in building production-ready AI and web projects, strengthening the club's technical presence at university and national levels.

Personal Projects

ComplaintHub - V2V Voice Pipeline for Citizen Grievance Automation

 [Link](#)

- Built a fully automated V2V (voice-to-voice) citizen complaint registration system where users interact entirely through natural speech over live call - no human agent involved. The pipeline ingests live audio, transcribes via ASR, reasons over an LLM, and delivers a synthesized spoken response, completing the full complaint lifecycle end-to-end.
- Engineered the real-time audio stack using GPT-4o-mini-realtime-preview model streaming for telephony (Twilio/Exotel), with custom G.711 μ -law converters, jitter buffer tuning, RTP optimization, and FFmpeg preprocessing.
- Integrated and did POC on three voice pipelines - OpenAI GPT-4o Realtime (PCM16 WebSocket protocol), Gemini Native Audio Thinking Model, and a custom STT-LLM-TTS micro-pipeline (Whisper → Groq gpt OSS → Google Wavenet) - with token-aware routing and latency-based model swapping to optimize cost and quality.
- Integrated Silero VAD with server-side threshold tuning and DeepFilterNet for real-time noise suppression, improving speech clarity and eliminating false-trigger interruptions in noisy call environments.
- Designed multi-stage NLP extraction to parse citizen intent, complaint category, and brand details from unstructured voice input, persisting structured records to Redis with async PostgreSQL for durable storage.

Syntropy Labs - LLM Evaluation & Benchmarking Platform

 [Link](#)

- Architected a LLM evaluation platform with four containerized microservices: an **Orchestrator** as the central API gateway for JWT auth, organization/project management, dataset lifecycle, and job orchestration; a **Model Runner** (FastAPI + LiteLLM) as a unified multi-provider LLM proxy; an **Eval Engine** (FastAPI) for automated metric scoring; and MongoDB for metadata persistence.
- Built the Model Runner as a provider-agnostic inference proxy using LiteLLM, normalizing API calls across OpenAI, Anthropic, Gemini, Mistral, and Azure into a single OpenAI-compatible interface with consistent token usage accounting and latency tracking per provider.
- Extended the Model Runner with a WebSocket-based Realtime Service supporting live voice evaluation integrating OpenAI Realtime API and Gemini BidiGenerateContent enabling audio-in/audio-out benchmarking models within the evaluation platform.
- Built the Eval Engine supporting statistical metrics, predefined LLM-as-judge metrics (relevance, groundedness, coherence, fluency), and fully custom evaluation criteria each configurable with a judge model, scoring threshold,

and per-metric weight for composite pass/fail scoring.

- Implemented multimodal evaluation pipelines for AI-generated image and video outputs scoring across structured JSON dimensions (physics plausibility, anatomical correctness, semantic adherence, temporal consistency, aesthetic quality) using vision-capable LLMs as structured evaluators.
- Designed a progressive batch evaluation system where the Orchestrator persists per-row scores back to google cloud storage hosted CSV datasets after every row, enabling real-time job progress polling from the frontend and partial result recovery on mid-job failures.

Aura.ai - Text-to-Video AI SaaS Platform



- Built an AI SaaS platform that automates end-to-end video creation from text/document inputs - covering LLM-powered content summarization, multilingual voiceover synthesis, sign language video generation, and AR/VR scene integration targeting accessibility in education and corporate training.
- Developed AuraBot, a knowledge-base-grounded assistant using a RAG pipeline for contextual Q&A over uploaded course or training material.
- Integrated an analytics dashboard tracking viewer engagement, quiz completion rates, comprehension scores, and retention metrics, using this data to personalize content delivery per user learning profile.

Education

Bachelor of Computer Applications - Information Technology

Chitkara University, Punjab, India (CGPA : 9.12)

Achievements

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- **Top 22 National Finalist & Social Buzz Winner** at Bajaj Finserv HackRx 5.0
 - **Top 50 Finalist** at Cars24 Token'26 hosted by OpenAI, AWS & elevenlabs
 - **Winner** at Aditya Birla Group's Synaptix Hackathon'25, hosted by Delhi Technological University
 - **Selected** for GitHub Field Day'24 at Microsoft, Gurgaon
 - **First Runner-Up** at FusionFest Hackathon, Chitkara University
 - **Finalist & Social Buzz Winner** at Vihaan 7.0, Delhi Technological University